第十九讲

组合变形

作业：10-2,8,11,16,20,21,22*

第十章　组合变形

10.1 弯拉（压）组合

10.2 弯扭组合

10.3 矩形截面杆组合变形一般情况

第十章 组合变形

10.1 弯拉（压）组合

1、组合变形分析基本思路

2、弯拉（压）组合

3、偏心压缩与截面核心

4、例题

1、组合变形分析基本思路

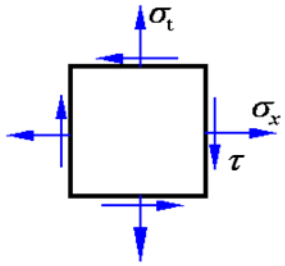
- 组合变形：两种以上基本变形的组合；
- 弯拉（压）组合（偏心压缩）、弯扭、拉弯扭、弯弯（双对称弯曲）；
- 薄壁筒：内（外）压 + 拉压弯扭。

基本变形危险截面应力分布

叠加

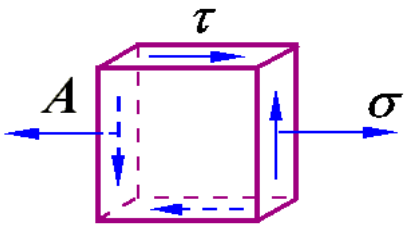
组合变形危险截面应力分布

危险点应力状态 → 强度条件



$$\sigma_x = \sigma_x^N + \sigma_x^{M_y} + \sigma_x^{M_z} + \sigma_x^p$$

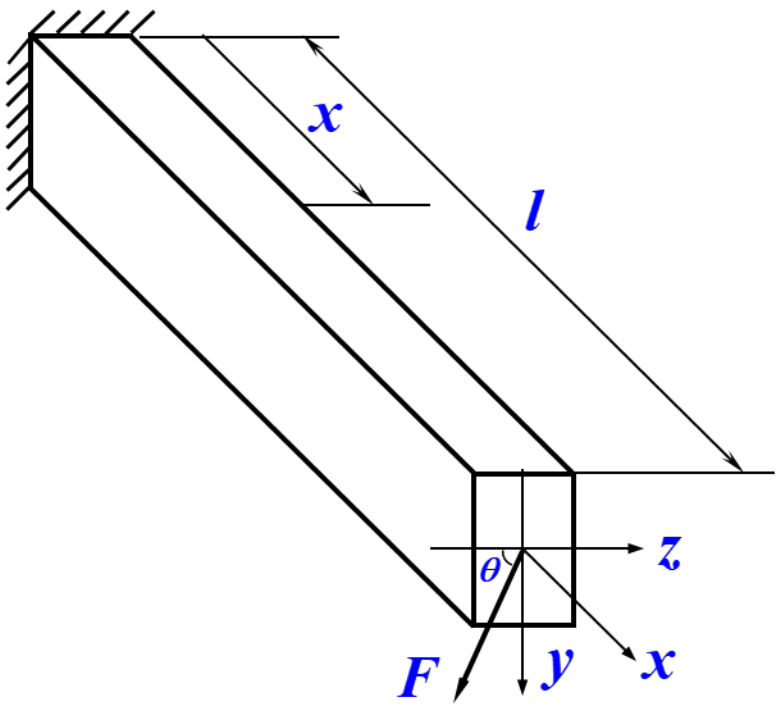
$$\sigma_t = \sigma_t^p, \quad \tau = \tau^T + \tau^{F_S}$$



$$\sigma_{r3} = \sqrt{\sigma^2 + 4\tau^2} \le [\sigma]$$

$$\sigma_{r4} = \sqrt{\sigma^2 + 3\tau^2} \le [\sigma]$$

1、组合变形分析基本思路

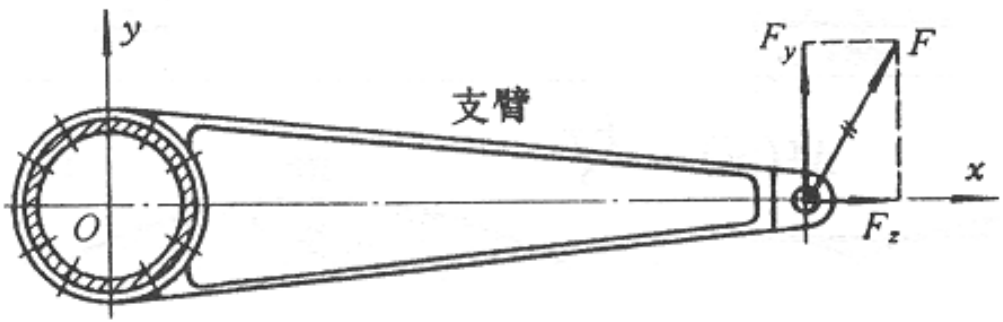


双对称弯曲——弯弯组合

$$\sigma(y,z)=\frac{M_y}{I_y}z-\frac{M_z}{I_z}y$$

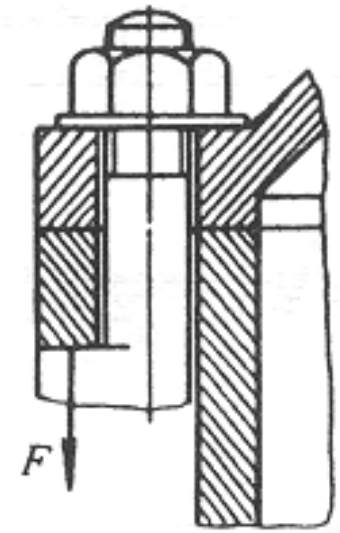
2、弯拉（压）组合

实例



弯拉组合

（横向载荷＋轴向载荷）

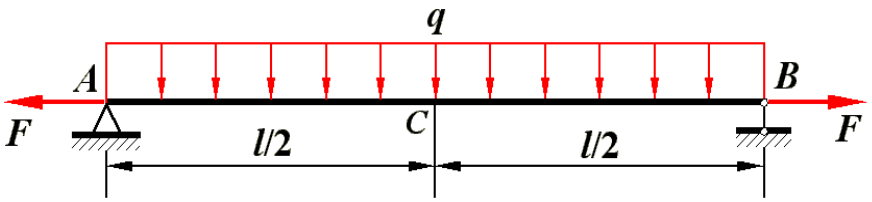


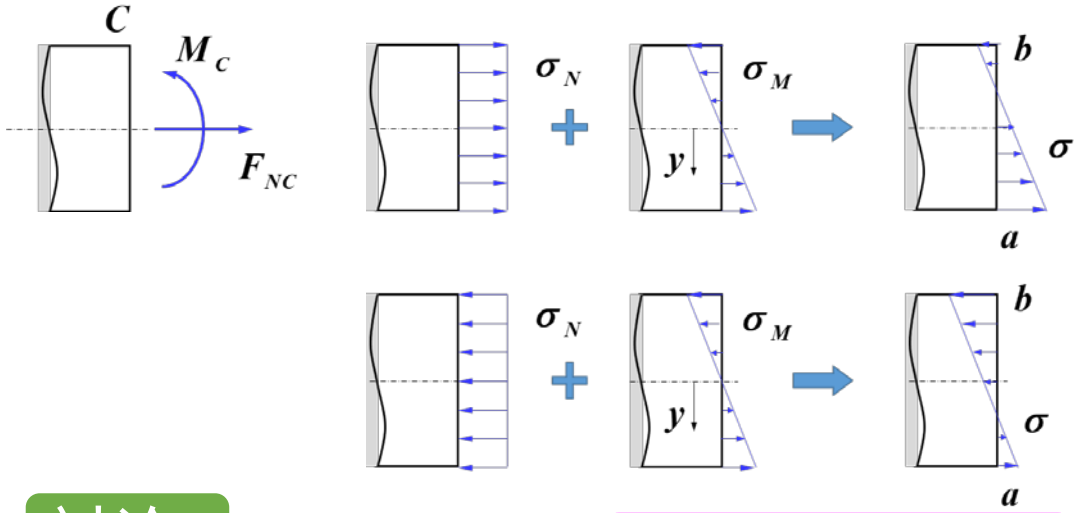
偏心拉伸

（外力平行于轴线）

2、弯拉（压）组合

图示简支梁，受轴向拉力、横向均布力同时作用





危险截面：C截面

$$F_{NC}=F,\quad M_{max}=M_C=\frac{ql^2}{8}$$

$$\sigma_N=\frac{F}{A},\quad \sigma_M=\frac{M_{max}}{I_z}y$$

$$\sigma=\sigma_N+\sigma_M=\frac{F}{A}+\frac{M_{max}}{I_z}y$$

危险点

$$\sigma_{max}=\sigma_a=\frac{F}{A}+\frac{M_{max}}{W_z}\le[\sigma]$$

讨论：

依据应力分布图
具体问题具体分析

1）若轴向受压

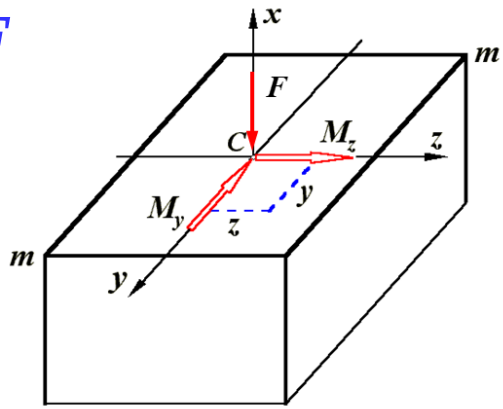
$$\sigma=-\frac{F}{A}+\frac{M_{max}}{I_z}y \qquad \sigma_{max}=?$$

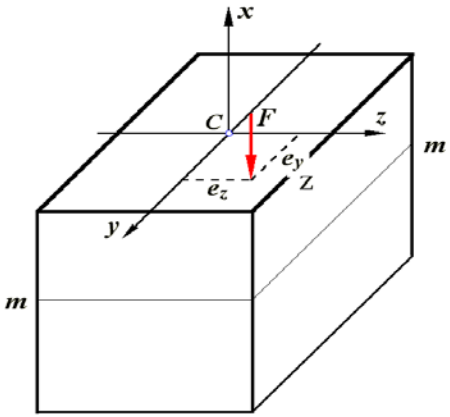
2）若材料为脆性

$$\sigma_{t,max}=? \quad \sigma_{c,max}=?$$

3、偏心压缩与截面核心

矩形截面柱，受偏心压力 F





应力分布

$$M_y = Fe_z, \quad M_z = Fe_y$$

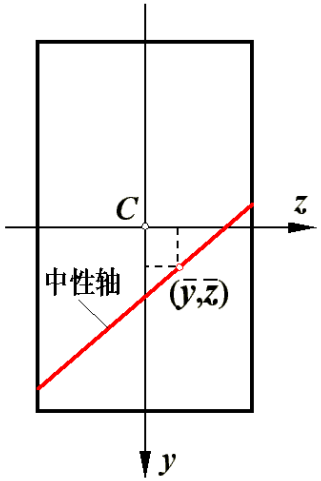
$$\sigma = -\frac{F}{A} - \frac{Fe_z z}{I_y} - \frac{Fe_y y}{I_z}$$

中性轴方程

$$\boxed{\frac{1}{A} + \frac{e_z \bar{z}}{I_y} + \frac{e_y \bar{y}}{I_z} = 0}$$

若 $\bar{z}=0$, $\quad \bar{y} = -\dfrac{I_z}{e_y A} = -\dfrac{h^2}{12e_y}$

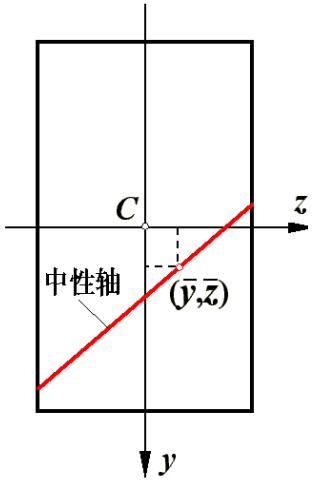
偏心距大，中性轴在面内，
截面应力有拉有压；
偏心距小，中性轴在面外，
截面应力有压无拉；

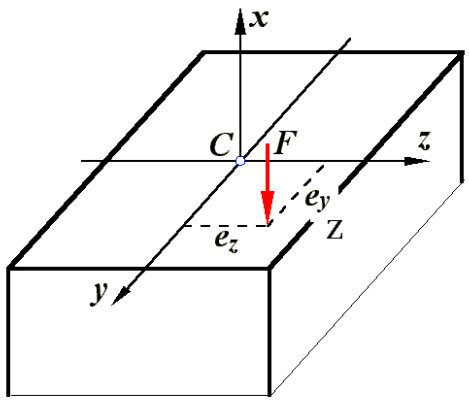


3、偏心压缩与截面核心

截面核心

$$\frac{1}{A}+\frac{e_z\bar{z}}{I_y}+\frac{e_y\bar{y}}{I_z}=0$$

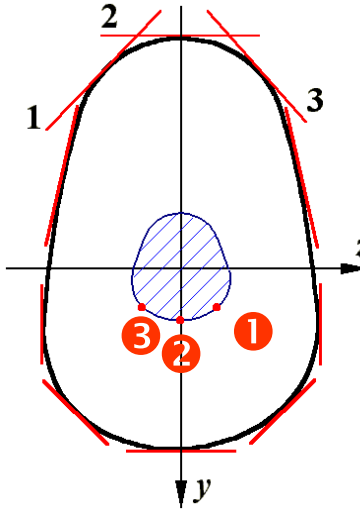




偏心距大，中性轴在面内，
截面应力有拉有压；
偏心距小，中性轴在面外，
截面应力有压无拉；

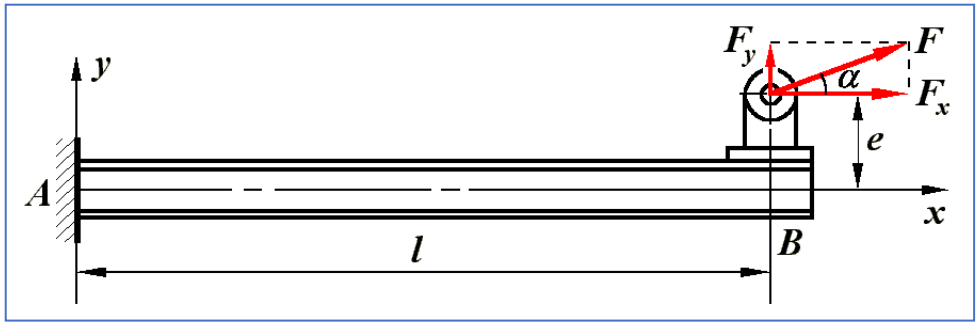
形心周围有一封闭区域，
偏心压力作用于其内，
截面不受拉——截面核心

混凝土柱偏心承压，
载荷宜在截面核心内。



4、例题

例10-1 图示工字梁，$F = 10\,\text{kN}$，$l = 2\,\text{m}$，$e = l/10$，$\alpha = 30°$，$[\sigma] = 160\,\text{MPa}$，选择工字钢型号。



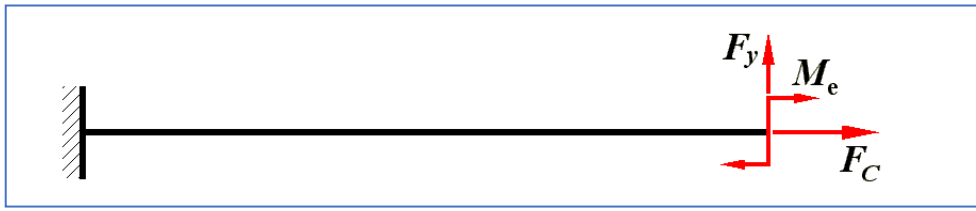
解： 1）外力分析

$$F_C = F_x = F\cos 30^\circ$$

$$F_y = F\sin 30^\circ$$

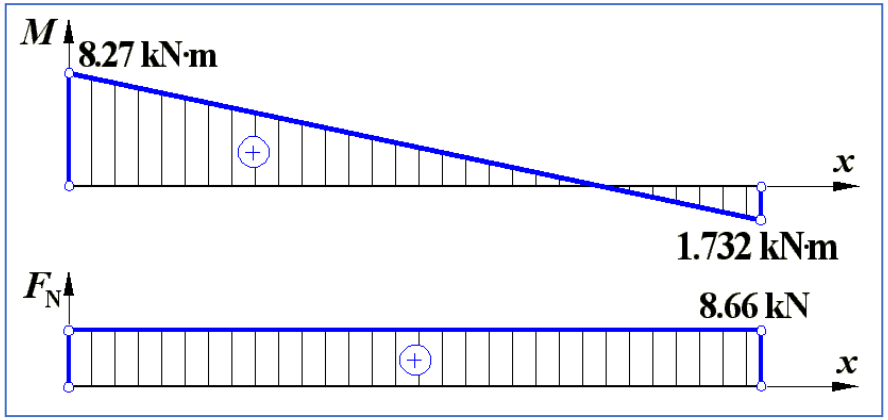
$$M_e = eF\cos 30^\circ$$

弯拉组合



2）应力分析

画QM图 ➡ 危险截面：A



强度条件：$$\frac{F_{NA}}{A} + \frac{M_A}{W_z} \le [\sigma]$$

4、例题

例10-1 图示工字梁，$F = 10\,\text{kN}$，$l = 2\,\text{m}$，$e = l/10$，$\alpha = 30°$，$[\sigma] = 160\,\text{MPa}$，选择工字钢型号。

解：

$$\frac{F_{NA}}{A} + \frac{M_A}{W_z} \le [\sigma]$$

3）按弯曲强度初步设计

$$\frac{M_A}{W_z} \le [\sigma] \qquad W_z \ge \frac{M_A}{[\sigma]} = 5.17 \times 10^{-5}\ \text{m}^4$$

选 №12.6，$W_z = 7.75 \times 10^{-4}\ \text{m}^4$，$A = 1.81 \times 10^{-3}\ \text{m}^2$

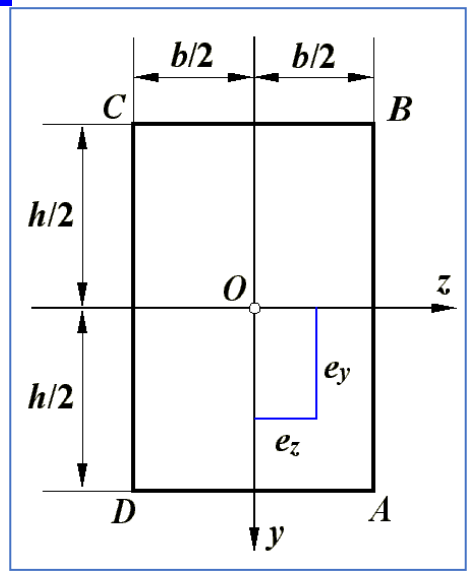
4）<u>校核与修改设计</u>

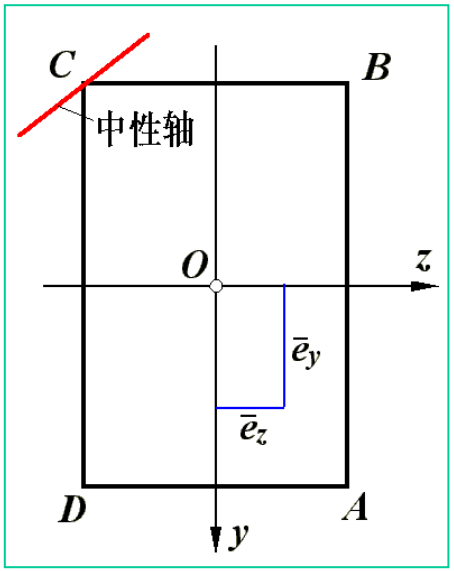
$$\sigma_{max} = \frac{F_{NA}}{A} + \frac{M_A}{W_z} = 111.5\ \text{MPa} \le [\sigma]$$

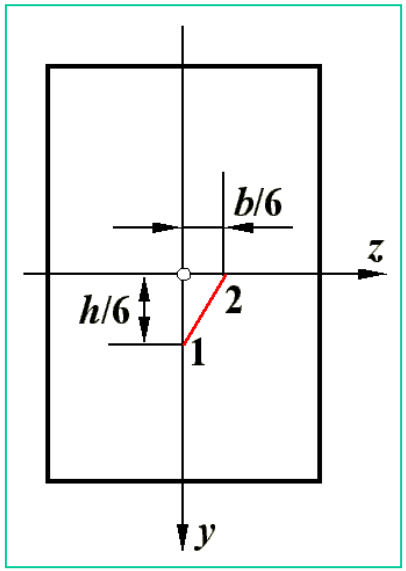
№12.6 满足强度要求，否则修改设计

4、例题

例10-2 确定矩形截面的截面核心。







解： 1. 轴压(e_z,e_y)在第1象限内，角点A受压，角点C应力

$$\sigma_C = -\frac{F}{bh} + \frac{6Fe_z}{hb^2} + \frac{6Fe_y}{bh^2}$$

偏心引起C点拉应力

轴压靠近形心，拉应力变小；
轴压远离形心，拉应力变大。

$\sigma_C = 0$

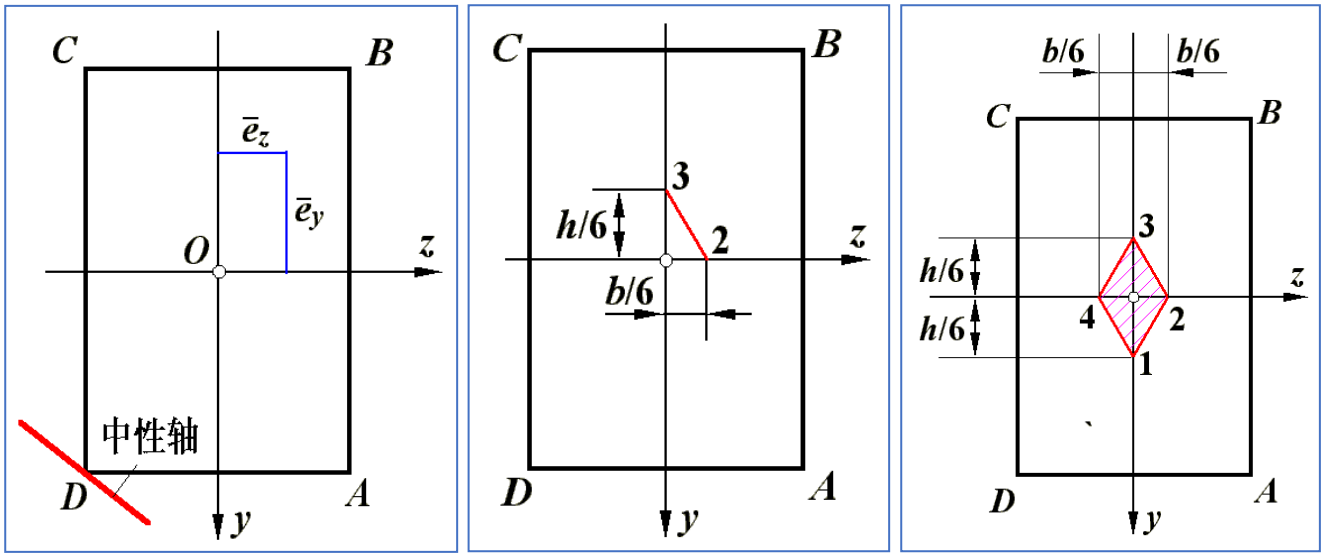
$$\frac{6\bar{e}_z}{b} + \frac{6\bar{e}_y}{h} = 1$$

第1象限内截面核心之边界

4、例题

例10-2 确定矩形截面的截面核心。

2. 其余象限内截面核心之边界



$$\sigma_D = \boxed{-\frac{F}{bh} + \frac{6Fe_z}{hb^2} - \frac{6Fe_y}{bh^2} = 0}$$

$$\frac{6\bar{e}_z}{b} - \frac{6\bar{e}_y}{h} = 1$$ —边界2-3

同理：得边界 3-4 与 4-1

第十章 组合变形

10.2 弯扭组合

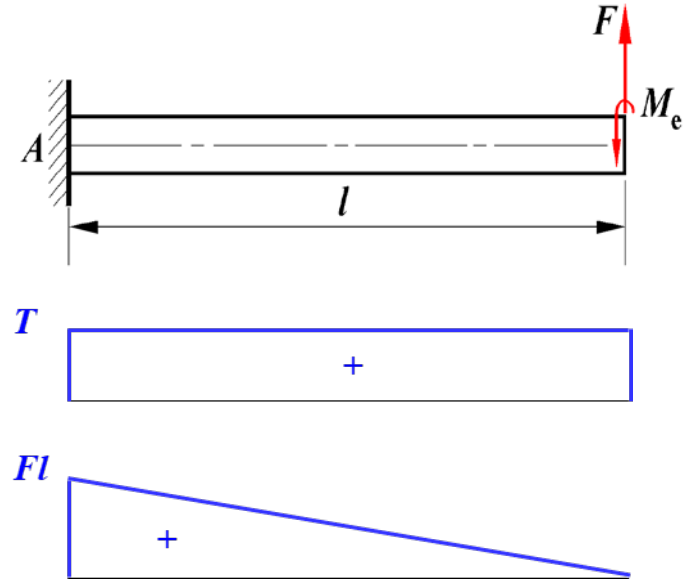
1、弯扭组合强度计算

2、弯拉（压）扭组合

3、例题

1、弯扭组合强度计算

图示圆截面悬臂梁，自由端受横向力和扭力矩



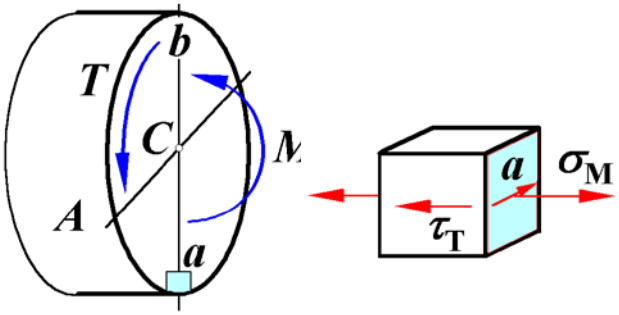
危险截面：A截面　　$M = Fl,\quad T = M_e$

危险点：a, b　　$\sigma_M = \dfrac{M}{W},\quad \tau_T = \dfrac{T}{W_P} = \dfrac{T}{2W}$

强度条件（塑性材料）

$$\sigma_{r3} = \sqrt{\sigma_M^2 + 4\tau_T^2} \le [\sigma]$$

$$\sigma_{r4} = \sqrt{\sigma_M^2 + 3\tau_T^2} \le [\sigma]$$



圆截面

$$\sigma_{r3} = \frac{\sqrt{M^2 + T^2}}{W} \le [\sigma] \qquad \sigma_{r4} = \frac{\sqrt{M^2 + 0.75T^2}}{W} \le [\sigma]$$

1、弯扭组合强度条件

强度条件的几种形式

（以第三、第四强度理论为例）

以主应力表示

$$\sigma_{r3}=\sigma_1-\sigma_3\le[\sigma]$$

为什么与单向拉伸比较？

简单、易行，给出材料性能

$$\sigma_{r4}=\frac{1}{\sqrt{2}}\sqrt{(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2}\le[\sigma]$$

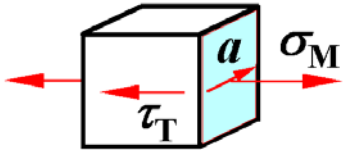
一般应力状态

以工作应力表示

$$\sigma_{r3}=\sqrt{\sigma_M^2+4\tau_T^2}\le[\sigma]$$

$$\sigma_{r4}=\sqrt{\sigma_M^2+3\tau_T^2}\le[\sigma]$$

常见平面应力状态



以内力表示

$$\sigma_{r3}=\frac{\sqrt{M^2+T^2}}{W}\le[\sigma]$$

$$\sigma_{r4}=\frac{\sqrt{M^2+0.75T^2}}{W}\le[\sigma]$$

塑性材料，圆截面轴，弯扭组合

形式变简单，适用范围变窄

2、弯拉（压）扭组合强度条件

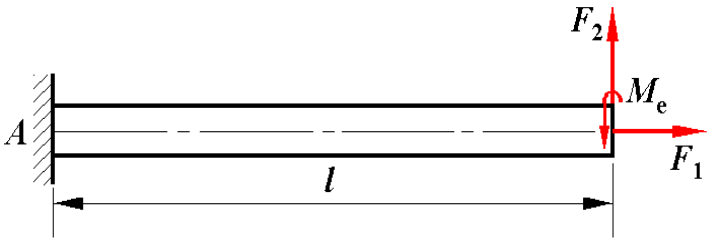
弯拉（压）扭组合

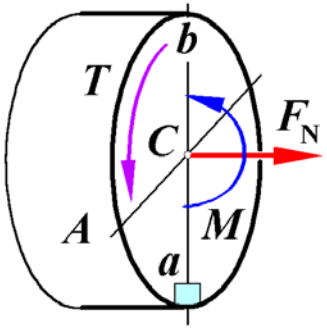
危险截面—截面A

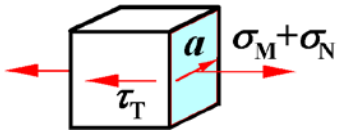
危 险 点— a

$$\sigma_a=\sigma_M+\sigma_N=\frac{M}{W}+\frac{F_N}{A}$$

$$\tau_a=\tau_T=\frac{T}{W_p}=\frac{T}{2W}$$







应力状态—常见平面应力状态

强度条件（塑性材料）

$$\sigma_{r3}=\sqrt{(\sigma_M+\sigma_N)^2+4\tau_T^2}\le[\sigma]$$

$$\sigma_{r4}=\sqrt{(\sigma_M+\sigma_N)^2+3\tau_T^2}\le[\sigma]$$

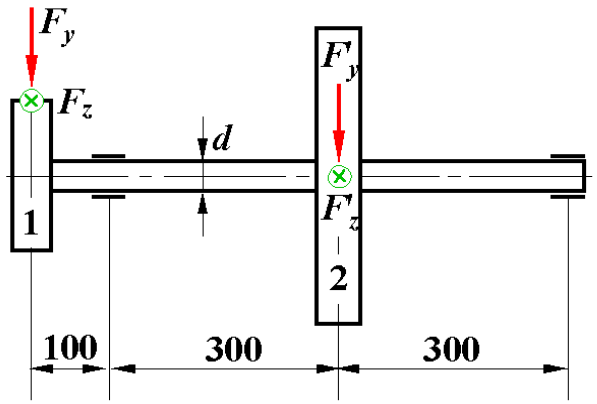
3、例题

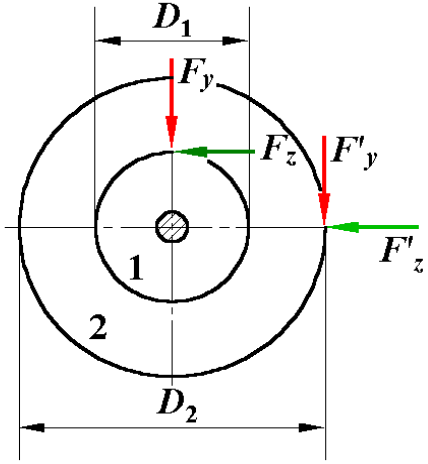
例10-3 图示钢质传动轴，F_y = 3.64 kN, F_z= 10 kN, F'_z =1.82 kN, F'_y = 5 kN, D_1 = 0.2 m, D_2 = 0.4 m, [σ] = 100 MPa, 轴径 d=52 mm. 试按第四强度理论校核轴的强度

解： 1）外力分析

$$M_1 = \frac{F_z D_1}{2} =$$

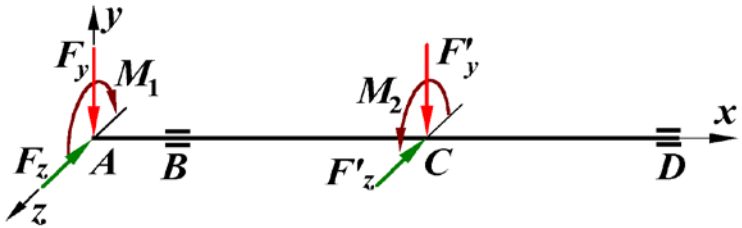
$$M_2 = \frac{F'_y D_2}{2} = 1\ \text{kN}\cdot\text{m}$$





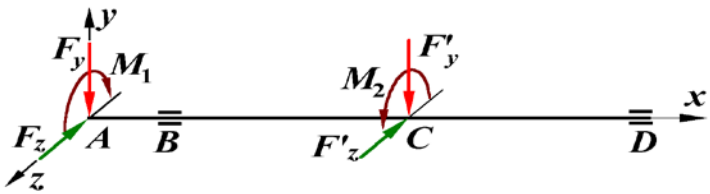
扭力矩

水平、铅垂横向力



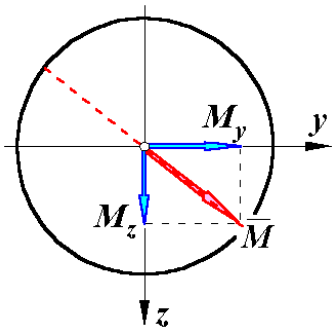
3、例题

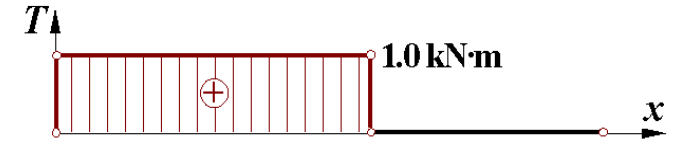
解： 2）画扭矩图、弯矩图

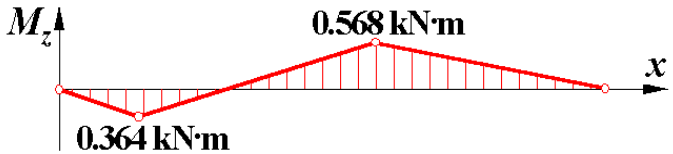


$$\overline{M}=\sqrt{M_y^2+M_z^2}$$

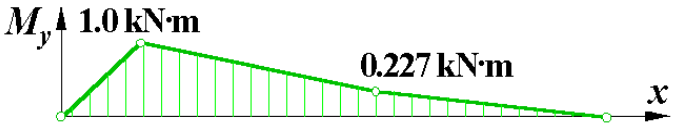
扭矩与合弯矩组合作用



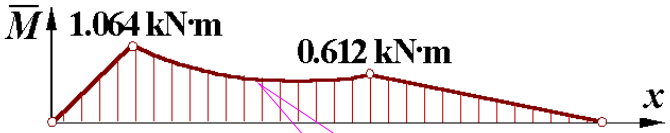




危险截面：*B*截面 ← *BC*段是凹曲线



3）强度校核



$$T_B=1.0\ \text{kN}\cdot\text{m} \qquad \overline{M}_B=1.064\ \text{kN}\cdot\text{m}$$

$$\sigma_{r4}=\frac{\sqrt{\overline{M}_B^2+0.75T_B^2}}{W}=99.4\ \text{MPa}\le[\sigma]$$

强度合格

证明见
习题10-12

第十章 组合变形

10.3 矩形截面杆组合变形一般情况

1、内力分析

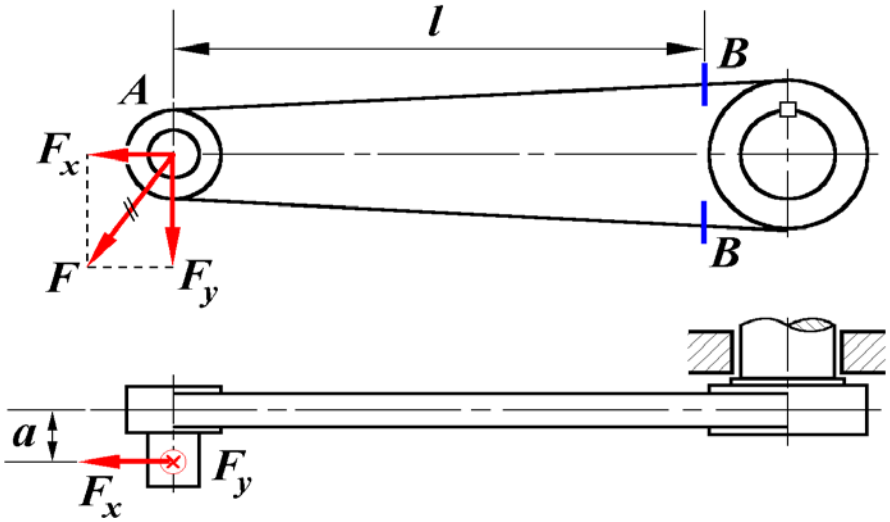
2、应力分析

3、强度条件

4、盒式梁弯扭组合

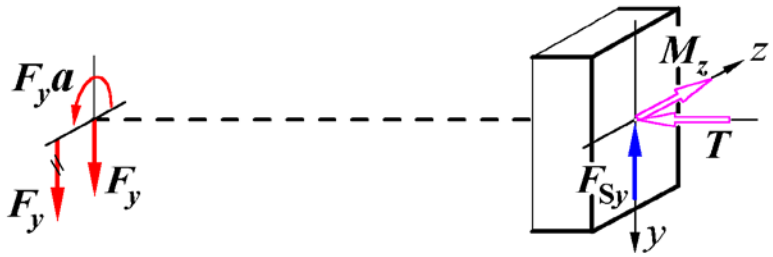
1、内力分析

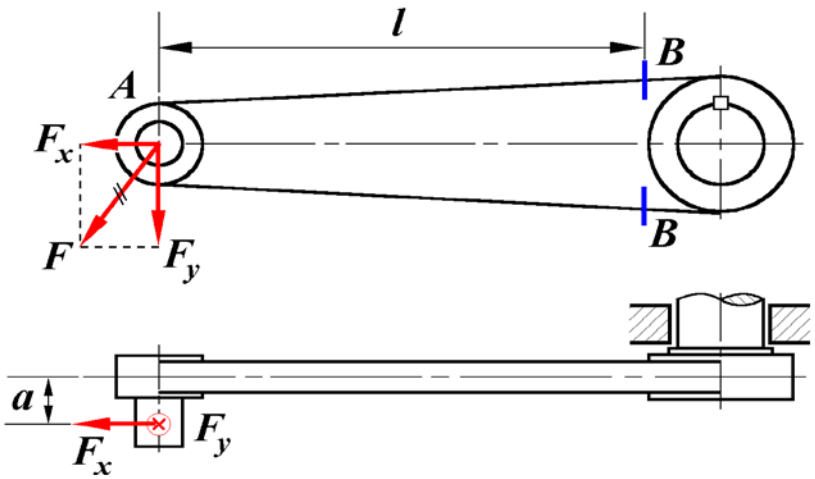
图示矩形截面钢质曲柄，受力F_x，F_y，建立B截面强度条件



1、内力分析

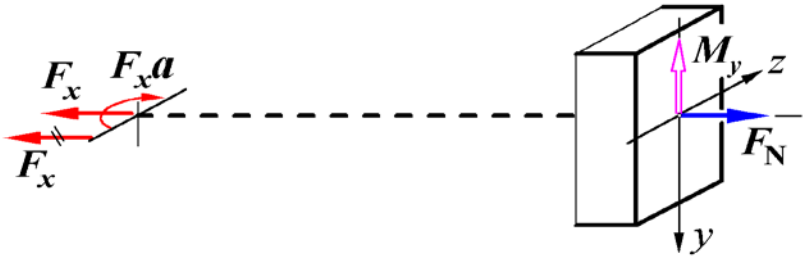
F_y引起的内力





$$F_{Sy}=F_y \quad M_z=F_y l \quad T=F_y a$$

F_x引起的内力



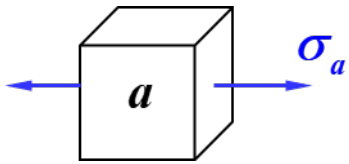
$$F_N=F_x$$

$$M_y=F_x a$$

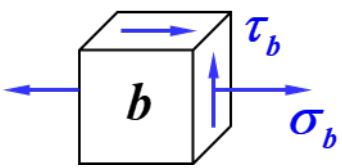
B截面有5个内力分量

2、应力分析

危险点：a, b, c

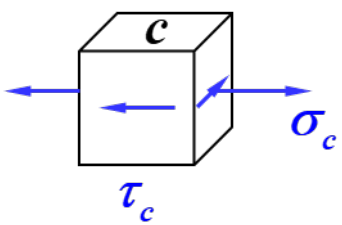


$$\sigma_a = \frac{M_y}{W_y} + \frac{M_z}{W_z} + \frac{F_N}{A} = \sigma_{max}$$



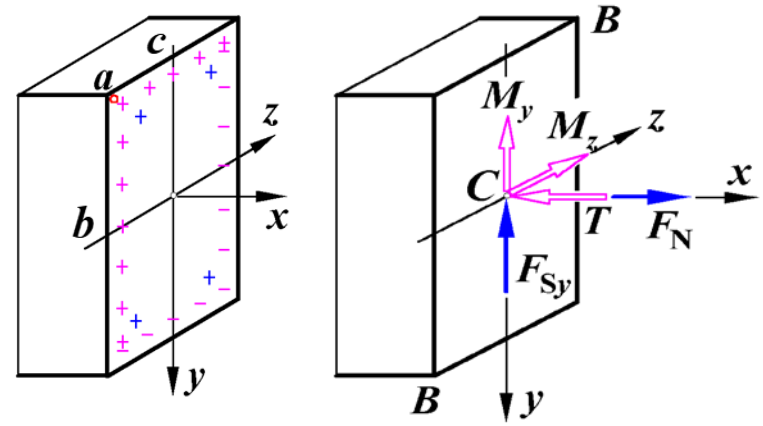
$$\sigma_b = \frac{M_y}{W_y} + \frac{F_N}{A}$$

$$\tau_b = \tau_b^T + \tau_b^S = \frac{T}{W_t} + \frac{3F_{Sy}}{2A}$$

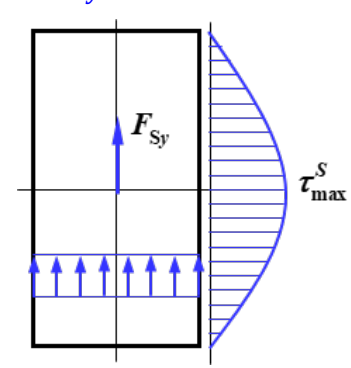


$$\sigma_c = \frac{M_z}{W_z} + \frac{F_N}{A} \qquad \tau_c = \frac{\gamma T}{W_t}$$

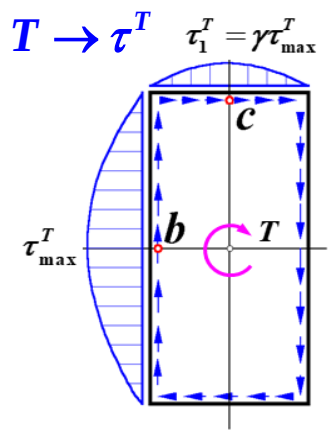
$M_y, M_z, F_N \rightarrow \sigma$



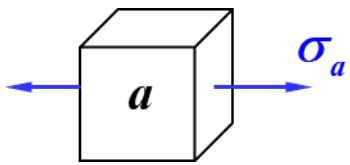
$F_{Sy} \rightarrow \tau^S$



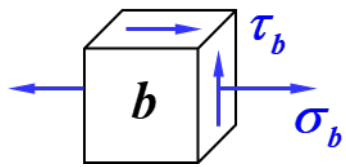
$T \rightarrow \tau^T$



3、强度条件 （以第三强度理论为例）

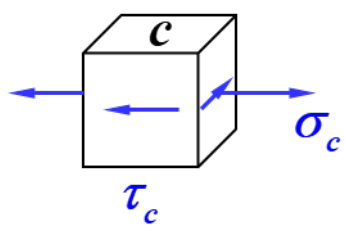


$$\sigma_a = \frac{M_y}{W_y} + \frac{M_z}{W_z} + \frac{F_N}{A} = \sigma_{max} \le [\sigma]$$



$$\sigma_{r3} = \sqrt{\left(\frac{M_y}{W_y} + \frac{F_N}{A}\right)^2 + 4\left(\frac{T}{W_t}\right)^2} \le [\sigma]$$

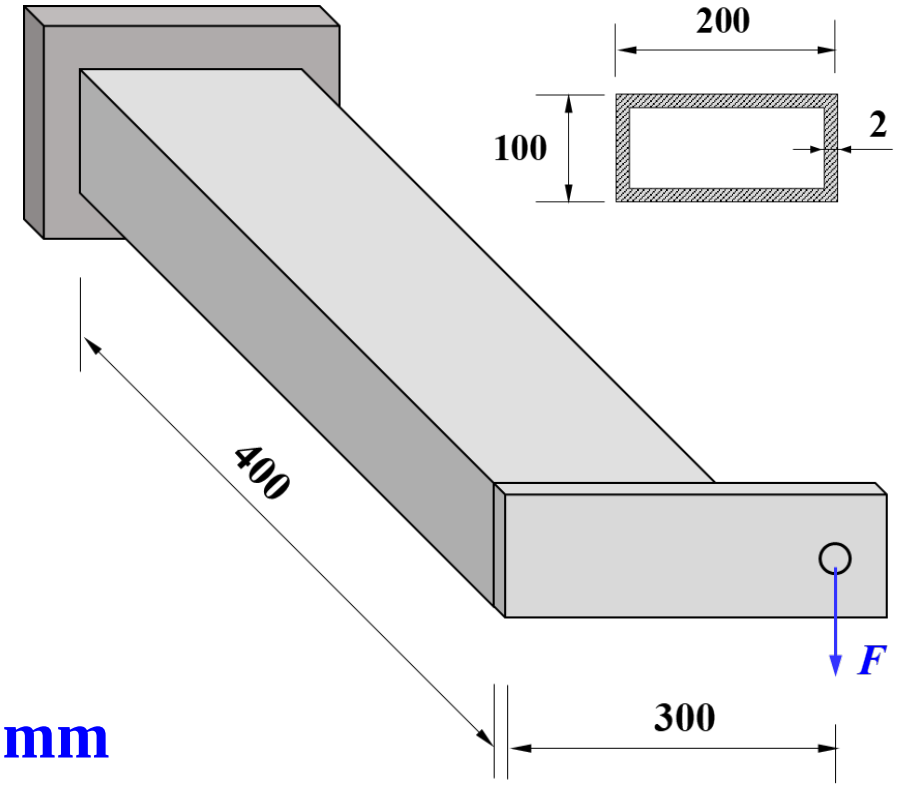
$$\sigma_{r3} = \sqrt{\left(\frac{M_y}{W_y} + \frac{F_N}{A}\right)^2 + 4\left(\frac{T}{W_t} + \frac{3F_S}{2A}\right)^2} \le [\sigma]$$



$$\sigma_{r3} = \sqrt{\left(\frac{M_z}{W_z} + \frac{F_N}{A}\right)^2 + 4\left(\frac{\gamma T}{W_t}\right)^2} \le [\sigma]$$

4、盒式梁弯扭组合变形

图示一端固定盒式梁，自由端焊接一刚性板，受偏心载荷 $F = 10$ kN，求σ_{r4}.



危险截面：根部截面

$$M = 400F = 4\times10^6\ \mathrm{N\cdot mm}$$

$$F_S = F = 1\times10^4\ \mathrm{N}$$

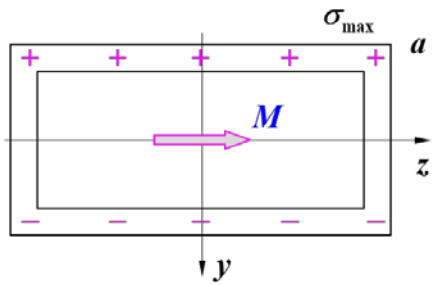
$$T = \left(300 - \frac{200}{2}\right)F = 2\times10^6\ \mathrm{N\cdot mm}$$

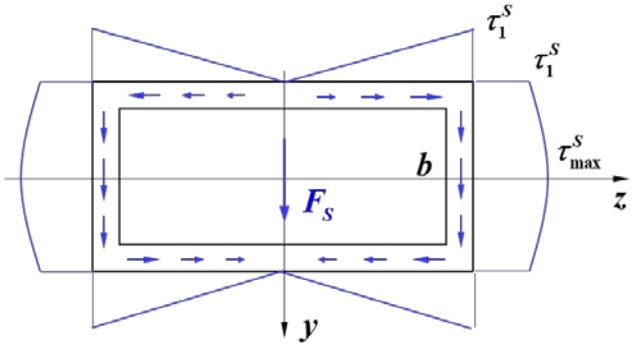
$$I_z = \frac{200}{12}\times100^3 - \frac{196}{12}\times96^3 = 2.216\times10^6\ \mathrm{mm}^4$$

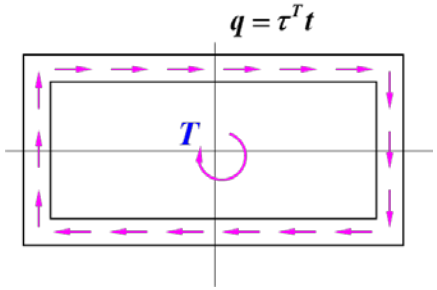
4、盒式梁弯扭组合变形

应力分布

画好应力分布图







$$\sigma_{max}=\frac{M}{I_z}\times 50=90.3\ \text{MPa}$$

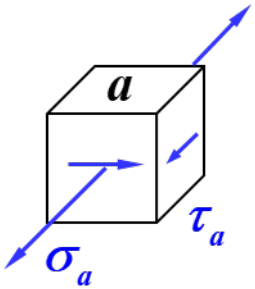
$$\tau_{max}^S=\frac{F_S S_z(\omega)}{I_z t}=27.3\ \text{MPa},$$

$$\tau_1^S=22.1\ \text{MPa}$$

$$\tau^T=\frac{T}{2\Omega t}=25.8\ \text{MPa}$$

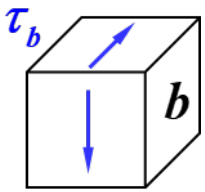
危险点

弯曲切应力足够大，不可忽略。



$$\sigma_a=\sigma_{max},\quad \tau_a=\tau_1^S+\tau^T$$

$$\sigma_{r4}=\sqrt{\sigma_a^2+3\tau_a^2}=122.6\ \text{MPa}$$



$$\tau_b=\tau_{max}^S+\tau^T$$

$$\sigma_{r4}=\sqrt{3}\tau_b=92\ \text{MPa}$$

第四强度理论相当应力为 122.6 MPa

本讲：组合变形

基本变形应力分布叠加

下讲：压杆稳定